

SHAMIT DHOLAKIYA

+1 (734) 891-1318 | dholakiashamit@gmail.com | linkedin.com/in/shamit30 | github.com/Shamit30

Summary

Entry-level Python developer with hands-on experience in backend services, REST APIs, and cloud-native applications. Skilled in developing, testing, and maintaining scalable Python systems. Strong foundation in software engineering principles, debugging, and Agile methodologies. Practical exposure to AI/ML and Generative AI.

Education

University of Michigan – Dearborn

Masters in Computer Information and Science, GPA: 4.0

Dearborn, MI

Jan 2025 – Present

Experience

Graduate Research Assistant

May 2025 – Present

University of Michigan – Dearborn

- Developed Python pipelines for acoustic and optical signal processing to support real-time anomaly detection.
- Built LSTM and ResNet models using modular design patterns for maintainable code.
- Applied Explainable AI methods (LIME, SHAP) and documented processes to increase transparency.
- Collaborated in Agile sprints, participated in code reviews, and contributed to technical documentation.
- **Skills:** Python, PyTorch, NumPy, Data Preprocessing, Signal Processing, Explainable AI, Agile

Software Engineer

Feb 2020 – Nov 2024

Seafflux Technologies

- Developed Python backend services and RESTful APIs for enterprise applications.
- Built real-time data pipelines with Kafka and Redis, improving throughput and reducing latency.
- Refactored legacy apps into Python microservices with FastAPI for better maintainability and deployment.
- Wrote unit and integration tests, reducing regression issues by 30%.
- Participated in Agile ceremonies and cross-team code reviews to ensure high-quality delivery.
- **Skills:** Python, FastAPI, PostgreSQL, Redis, Kafka, Docker, AWS, Git, Linux, Agile/Scrum

Projects

Campus Support Chatbot

- Built a Python chatbot using LangChain and FastAPI to provide real-time student support.
- Integrated APIs (Google Sheets, Tavily search) and ChromaDB for dynamic, context-aware responses.
- Implemented logging, error handling, and query rewriting for reliability.
- **Skills:** Python, FastAPI, LangChain, ChromaDB, REST APIs, Git, Cloud Deployment

Real-Time Asset Monitoring System

- Developed Python microservices for geofencing alerts and predictive monitoring.
- Integrated Redis queues and Elasticsearch for real-time analytics and anomaly detection.
- Containerized applications with Docker and deployed on Kubernetes using CI/CD pipelines.
- **Skills:** Python, Redis, Elasticsearch, Microservices, Docker, Kubernetes, CI/CD, Git

Nowcasting with Satellite Images

- Preprocessed 14K+ satellite images and built ConvLSTM-CNN models for spatiotemporal weather prediction.
- Reduced forecasting error by 25% and generated visual dashboards using Matplotlib.
- **Skills:** Python, TensorFlow, ConvLSTM, CNN, NumPy, Matplotlib, Data Preprocessing

Technical Skills

Programming: Python, JavaScript, TypeScript, Java

Frameworks & APIs: FastAPI, Flask, Django (exposure), Node.js, Express.js, REST APIs

Databases: PostgreSQL, Redis, Elasticsearch, MongoDB

Cloud/DevOps: AWS (Lambda, EC2, S3), GCP (Cloud Run), Docker, Kubernetes, CI/CD

AI/ML: CNN, LSTM, ConvLSTM, Predictive Analytics, Explainable AI (LIME, SHAP)

Generative AI: LangChain, LangGraph, Google Gemini, OpenAI GPT-4, RAG

Tools: Git, Linux, Agile/Scrum, Postman, Auth0